

SOFTWARE ENGINEERING

LAB - 1 Requirements Engineering & UML Use-Case Modelling

Name : Inchara K Kuppal

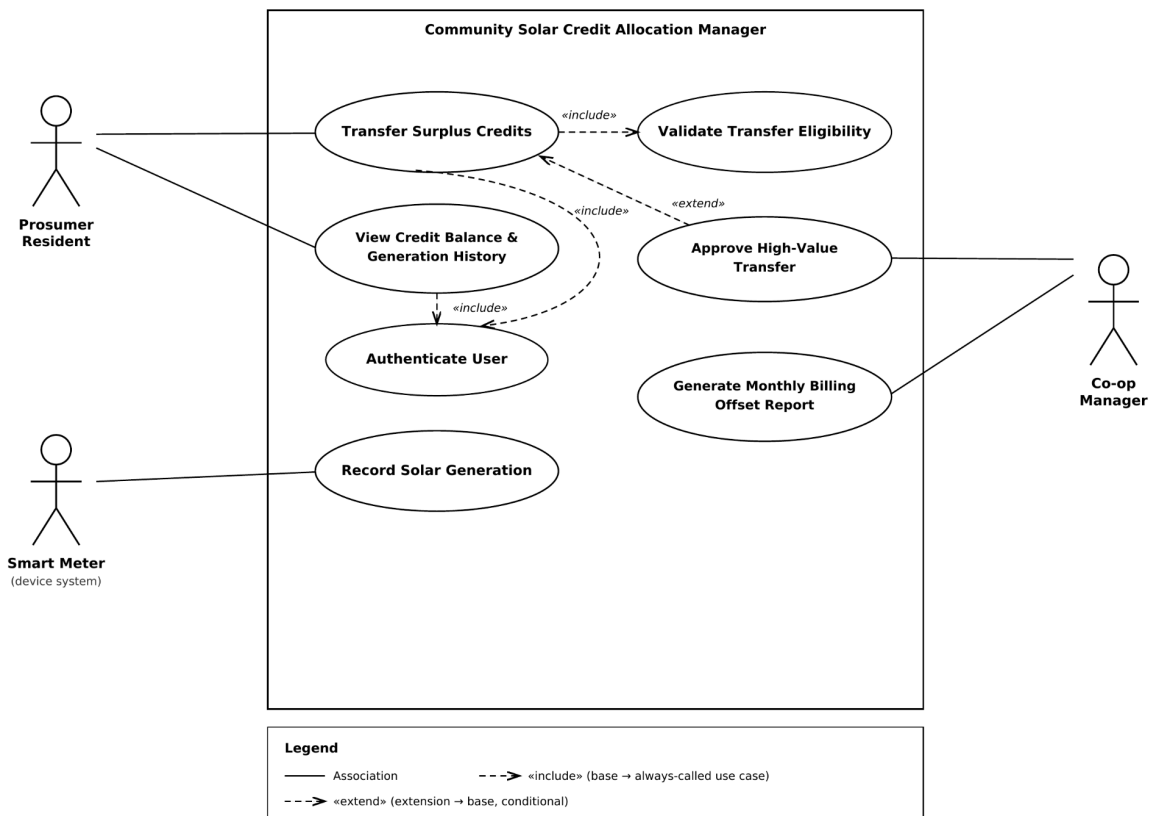
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PROBLEM STATEMENT #63 | Sustainability & Green Tech

Community Solar Credit Allocation Manager

Use-Case Diagram: Community Solar Credit Allocation Manager



Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record solar generation units (kWh) from smart meter feeds and allow prosumers to transfer surplus credits to neighbor accounts.	High	Pass: Transferred credits deduct from prosumer balance and apply to neighbor billing statement; Fail: Transfer exceeding balance.	Core requirement for peer-to-peer microgrid energy sharing.
FR-002	Functional	The system shall display a real-time dashboard showing the prosumer's current solar credit balance and generation history.	High	Pass: Dashboard updates within 1 minute of smart meter data ingestion.	Prosumers need visibility into their assets to make transfer decisions.

FR-003	Functional	The system shall generate a monthly report calculating the utility billing offsets for each prosumer based on their net credit balance.	Medium	Pass: System outputs a downloadable report on the 1st of each month with accurate offset calculations.	Necessary for financial tracking and integration with traditional utility billing.
FR-004	Functional	The system shall require the Co-op Manager to review and approve any credit transfers exceeding 500 kWh in a single transaction.	Low	Pass: Transfers > 500 kWh are held in "Pending" status until Manager clicks "Approve."	Prevents fraudulent or erroneous mass transfers within the grid.
FR-005	Functional	The system shall authenticate prosumers and managers before granting access to the portal.	High	Pass: Users providing correct credentials gain access; incorrect attempts yield an error.	Protects user financial and energy data from unauthorized access.

NFR-001	Nonfunctional	The smart meter telemetry processing pipeline must validate and store hourly generation data for 500 households with 99.9% uptime.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the system can handle concurrent IoT data streams without crashing.
NFR-002	Nonfunctional	The system shall encrypt all credit transfer transactions and billing offset data using AES-256 encryption.	High	Pass: Security audit verifies data at rest and in transit is encrypted.	Maintains data privacy and prevents tampering with financial-equivalent energy credits.

Use-Case Flow Specification: Transfer Surplus Credits

Use Case: UC-02 Transfer Surplus Credits

Actor: Prosumer Resident

Preconditions: The prosumer is authenticated and has a positive surplus credit balance.

Postconditions: The specified credit amount is deducted from the prosumer's balance and credited to the recipient's account.

Main Success Scenario:

1. The Prosumer Resident navigates to the "Transfer Credits" module in the portal.
2. The system retrieves and displays the user's current available surplus balance.
3. The Prosumer inputs the recipient's account ID and the amount of credits (kWh) to transfer.
4. The system validates that the entered amount does not exceed the available balance and that the recipient ID is valid (triggering UC-07 Validate Transfer Eligibility).
5. The Prosumer confirms the transaction details.
6. The system processes the transfer, updating both accounts in the backend.
7. The system displays a success message with a transaction receipt.
8. The use case ends successfully.

Alternate Flow 1: Insufficient Balance 4a. The system detects the entered transfer amount exceeds the available surplus balance.

- 4a1. The system highlights the input field in red and displays an "Insufficient Credits" error message.
- 4a2. The Prosumer is prompted to enter a lower amount or cancel the transaction.
- 4a3. If the user cancels, the system returns to the main dashboard.

Alternate Flow 2: High-Value Transfer Threshold Met 4b. The system detects the transfer amount exceeds the 500 kWh threshold.

- 4b1. The system triggers the extension use case (UC-04 Approve High-Value Transfer).
- 4b2. The system notifies the Prosumer that the transfer requires Co-op Manager approval and changes the transaction status to "Pending".